answering Leibniz's question of why there is something rather than nothing.

Can String Theory Be Experimentally Tested?

Among the many features of string theory that we have discussed in the preceding chapters, the following three are perhaps the most important ones to keep firmly in mind. First, gravity and quantum mechanics are part and parcel of how the universe works and therefore any purported unified theory must incorporate both. String theory accomplishes this. Second, studies by physicists over the past century have revealed that there are other key ideas—many of which have been experimentally confirmed—that appear central to our understanding of the universe. These include the concepts of spin, the family structure of matter particles, messenger particles, gauge symmetry, the equivalence principle, symmetry breaking, and supersymmetry, to name a few. All of these concepts emerge naturally from string theory. Third, unlike more conventional theories such as the standard model, which has 19 free parameters that can be adjusted to ensure agreement with experimental measurements, string theory has no adjustable parameters. In principle, its implications should be thoroughly definitive—they should provide an unambiguous test of whether the theory is right or wrong.

The road from this "in principle" ratiocination to an "in practice" fact is encumbered by many hurdles. In Chapter 9 we described some of the technical obstacles, such as determining the form of the extra dimensions, that currently stand in our way. In Chapters 12 and 13 we placed these and other obstacles in the broader context of our need for an exact understanding of string theory, which, as we have seen, naturally leads us to the consideration of M-theory. No doubt, achieving a full understanding of string/M-theory will require a great deal of hard work and an equal amount of ingenuity.

At every step of the way, string theorists have sought and will continue to seek experimentally observable consequences of the theory. We must not lose sight of the long-shot possibilities for finding evidence of string theory discussed in Chapter 9. Furthermore, as our understanding deepens there will, no doubt, be other rare processes or features of string theory that will suggest yet other indirect experimental signatures.